

Rebuilding the Social Engine — Technical Report

by Team Tekton – Yukta H, Chaaru Shree S

1. Problem Definition

The Social Engine's semantic understanding layer has failed, leaving the system unable to properly understand the meaning, tone, and intent of human language. Round 2 therefore focuses on rebuilding this NLP-driven comprehension layer using the provided labelled textual dataset. The system must use the available social-media text to recover semantic understanding through tasks such as sentiment and topic classification.

2. Objectives

- Rebuild the failed semantic understanding layer of the Social Engine using NLP techniques.
- Preprocess and transform social-media text into suitable numerical representations for machine learning.
- Perform sentiment classification into Negative, Neutral, and Positive categories.
- Perform topic classification across Account_Security, Community_Discussion, Feature_Feedback, and Technical_Issues.
- Compare lightweight machine learning and deep learning approaches using appropriate evaluation metrics.
- Analyse model errors and predictions to understand classification behaviour and improve the semantic understanding of the Social Engine.

3. Pre-processing Pipeline

- Checked the dataset for missing values, duplicate rows, duplicate texts, and empty or very short posts.
- Converted text to lowercase for consistent representation.
- Removed URLs and user mentions from the posts.
- Removed the # symbol while retaining the hashtag word.
- Removed special characters while preserving alphanumeric tokens such as some1, 1st, and 5G.
- Normalized extra whitespace and stripped leading/trailing spaces.
- Stopword removal was not applied to preserve potentially useful contextual information in short social-media text.
- Converted the cleaned text into numerical features using TF-IDF for machine learning models.

4. Model Selection

The following models were tested for sentiment and/or topic classification:

1. TF-IDF + Logistic Regression (Baseline)
2. Improved TF-IDF + Logistic Regression
3. VADER Sentiment
4. Multinomial Naive Bayes
5. Linear SVM
6. Simple Deep Learning
7. BiLSTM

Model	Task	Accuracy	Macro F1
TF-IDF + Logistic Regression (Baseline)	Sentiment	0.5712	0.5708
Improved TF-IDF + Logistic Regression	Sentiment	0.5734	0.5730
VADER	Sentiment	0.5819	0.5753
Multinomial Naive Bayes	Sentiment	0.5712	0.5653
Linear SVM	Sentiment	0.5447	0.5440
Simple Deep Learning	Sentiment	0.5478	0.5470
BiLSTM	Sentiment	0.5685	0.5693
TF-IDF + Logistic Regression (Baseline)	Topic	0.8828	0.3723
Improved TF-IDF + Logistic Regression	Topic	0.9273	0.6591
Multinomial Naive Bayes	Topic	0.8572	0.2308
Linear SVM	Topic	0.9349	0.6706
Simple Deep Learning	Topic	0.3754	0.2475
BiLSTM	Topic	0.8662	0.4577

The final pipeline uses TF-IDF + Logistic Regression for sentiment **and** TF-IDF + Linear SVM for topic classification.

VADER wasn't selected because it is a lexicon/rule-based method, not a trainable supervised model, so the final pipeline uses TF-IDF + Logistic Regression for sentiment despite VADER's slightly higher macro F1.

5. Training Methodology

- The dataset was divided into 75% training and 25% testing, resulting in 6,773 training samples and 2,227 test samples.
- A group-aware split based on cleaned text was used because the dataset contained repeated text entries. This ensured that identical texts did not appear in both training and test sets.
- TF-IDF was used to convert the cleaned social-media text into numerical feature representations for both final models.
- Logistic Regression was trained for sentiment classification across Negative, Neutral, and Positive classes.
- Linear SVM was trained for topic classification across Account_Security, Community_Discussion, Feature_Feedback, and Technical_Issues.
- Class weighting was used to address the strong imbalance in the topic classes, particularly the dominance of Community_Discussion.
- The test set was kept separate from model tuning and was used only for the final performance evaluation.
- The final TF-IDF vectorizer, Logistic Regression sentiment model, and Linear SVM topic model were saved together in the trained model file for future prediction and deployment.

6. Evaluation Metrics

The models were evaluated using Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Weighted F1-score.

6.1 Sentiment Classification — Model Performance

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
VADER	0.5819	0.5876	0.5809	0.5753	0.5762
Improved TF-IDF + Logistic Regression	0.5734	0.5731	0.5730	0.5730	0.5735
TF-IDF + Logistic Regression (Baseline)	0.5712	0.5710	0.5707	0.5708	0.5713
BiLSTM	0.5685	0.5746	0.5685	0.5693	0.5695
Multinomial Naive Bayes	0.5712	0.5660	0.5649	0.5653	0.5654
Simple Deep Learning	0.5478	0.5589	0.5482	0.5470	0.5471
Linear SVM	0.5447	0.5452	0.5436	0.5440	0.5442

6.2 Topic Classification — Model Performance

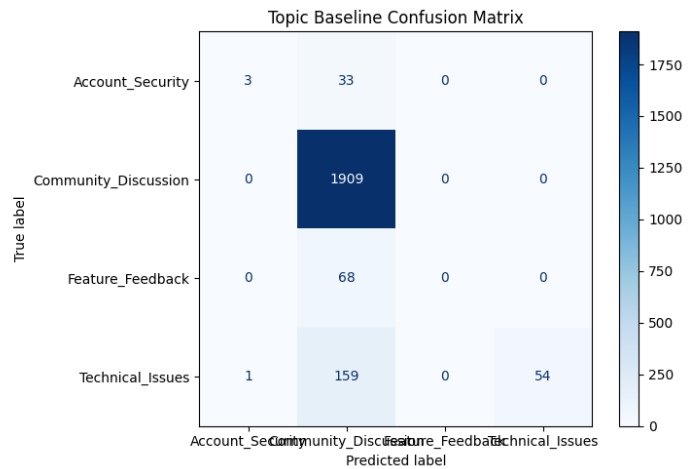
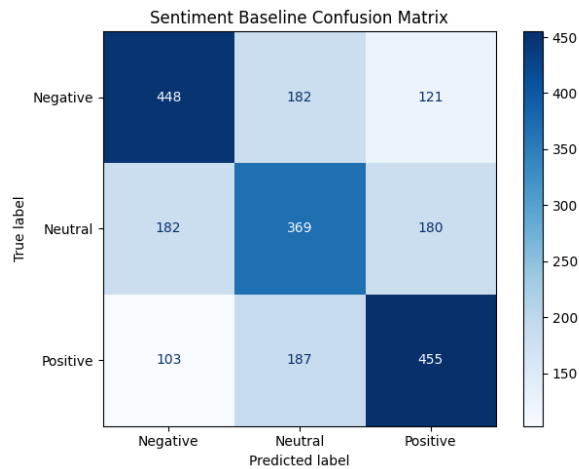
Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Linear SVM	0.9349	0.8875	0.5760	0.6706	0.9210
Improved TF-IDF + Logistic Regression	0.9273	0.8272	0.5796	0.6591	0.9187
BiLSTM	0.8662	0.4875	0.4550	0.4577	0.8585
TF-IDF + Logistic Regression (Baseline)	0.8828	0.6575	0.3339	0.3723	0.8437
Simple Deep Learning	0.3754	0.3118	0.4424	0.2475	0.4695
Multinomial Naive Bayes	0.8572	0.2150	0.2500	0.2308	0.7970

6.3 Metric Definitions

- Accuracy: Proportion of correctly classified samples.
- Macro Precision: Average precision across all classes with equal class importance.
- Macro Recall: Average recall across all classes with equal class importance.
- Macro F1: Harmonic mean of precision and recall calculated equally across classes.
- Weighted F1: F1-score weighted according to the number of samples in each class.
- Macro F1 is particularly important for topic classification because the topic classes are highly imbalanced.

7. Confusion Matrix

7.1 Baseline Confusion Matrices



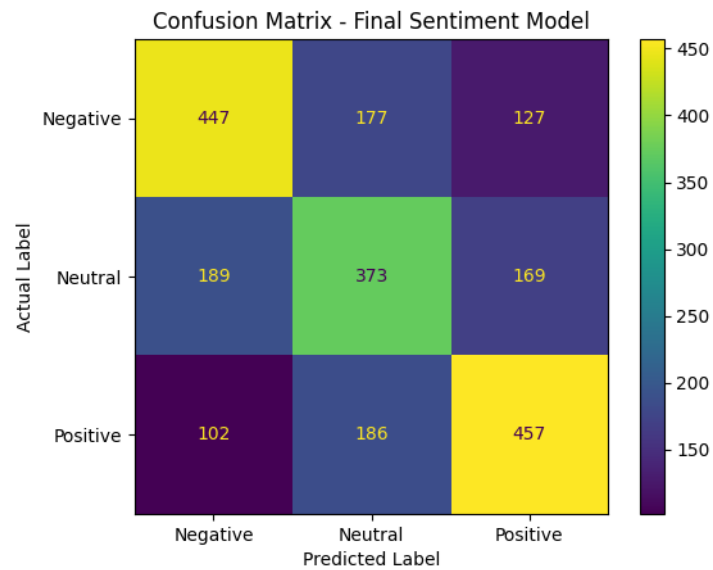
Sentiment Baseline — TF-IDF + Logistic Regression

- Negative: 448 correctly classified; 182 predicted as Neutral and 121 as Positive.
- Neutral: 369 correctly classified; 182 predicted as Negative and 180 as Positive.
- Positive: 455 correctly classified; 103 predicted as Negative and 187 as Neutral.
- Neutral was frequently confused with both Negative and Positive.

Topic Baseline — TF-IDF + Logistic Regression

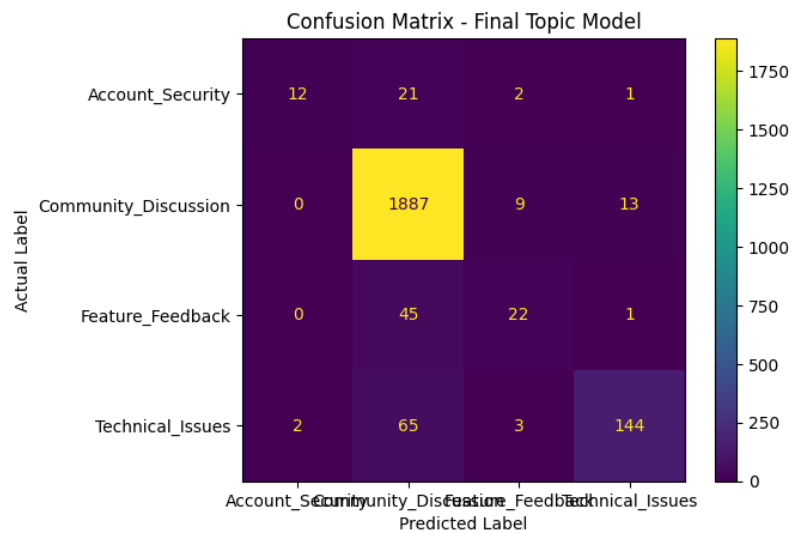
- Account_Security: 3 correctly classified; 33 predicted as Community_Discussion.
- Community_Discussion: 1,909 correctly classified.
- Feature_Feedback: 0 correctly classified; all 68 predicted as Community_Discussion.
- Technical_Issues: 54 correctly classified; 159 predicted as Community_Discussion.
- The baseline strongly predicted Community_Discussion, resulting in substantial misclassification of minority-topic samples.

7.2 Final Sentiment Model — TF-IDF + Logistic Regression



- Negative: 447 samples were correctly classified, while 177 were predicted as Neutral and 127 as Positive.
- Neutral: 373 samples were correctly classified, while 189 were predicted as Negative and 169 as Positive.
- Positive: 457 samples were correctly classified, while 102 were predicted as Negative and 186 as Neutral.
- The largest confusion pairs were Neutral → Negative (189), Positive → Neutral (186), and Negative → Neutral (177).
- The matrix shows that Neutral remains the main source of sentiment confusion.

7.3 Final Topic Model — TF-IDF + Linear SVM



- Account_Security: 12 samples were correctly classified; 21 were predicted as Community_Discussion.
- Community_Discussion: 1,887 samples were correctly classified; 9 were predicted as Feature_Feedback and 13 as Technical_Issues.
- Feature_Feedback: 22 samples were correctly classified; 45 were predicted as Community_Discussion.
- Technical_Issues: 144 samples were correctly classified; 65 were predicted as Community_Discussion.
- The largest topic error was Technical_Issues → Community_Discussion (65), followed by Feature_Feedback → Community_Discussion (45).
- The matrix shows that several minority-topic samples are still classified as the dominant Community_Discussion class.

8. Error Analysis and Explainability

Error analysis was performed using the predictions of the final selected models. The most frequent confusion patterns were identified, followed by feature-level analysis using the TF-IDF coefficients of the final Logistic Regression and Linear SVM models.

8.1 Sentiment Error Analysis

Actual	Predicted	Error Count
Neutral	Negative	189
Positive	Neutral	186
Negative	Neutral	177
Neutral	Positive	169
Negative	Positive	127
Positive	Negative	102

- The largest confusion was Neutral → Negative (189).
- Positive → Neutral (186) and Negative → Neutral (177) were the next largest errors.
- The error patterns show that Neutral is the main source of sentiment confusion.

8.2 Topic Error Analysis

Actual	Predicted	Error Count
Technical_Issues	Community_Discussion	65
Feature_Feedback	Community_Discussion	45
Account_Security	Community_Discussion	21
Community_Discussion	Technical_Issues	13
Community_Discussion	Feature_Feedback	9
Technical_Issues	Feature_Feedback	3
Account_Security	Feature_Feedback	2
Technical_Issues	Account_Security	2
Account_Security	Technical_Issues	1
Feature_Feedback	Technical_Issues	1

- The largest topic error was Technical_Issues → Community_Discussion (65).
- Feature_Feedback → Community_Discussion (45) and Account_Security → Community_Discussion (21) were also frequent.
- The errors show that minority-topic samples are frequently classified as the dominant Community_Discussion class.

8.3 Sentiment Explainability

Class	Top Supporting Features
Negative	back jordi, family for, cena vs, back saturday, be sept, declined, apartheid, cd on, export, end cliff
Neutral	be down, federal judge, chelsea and, attend the, better in, average, evolution of, died, 4th career, already do
Positive	building on, bayer leverkusen, be over, about power, be available, assumed, class just, 5th place, ever today, 2nd album

- The Logistic Regression model uses different TF-IDF features to distinguish the three sentiment classes.
- Several features appear to be dataset-specific, showing statistical associations learned from the training data.
- The feature patterns provide evidence for the observed difficulty in distinguishing Neutral from Negative and Positive.

8.4 Topic Explainability

Class	Top Supporting Features
Account_Security	7pm u002c, 1st rd, 7th grader, 7th, being ignored, done serious, 7pm, cak alcoa, be greeted, 7th harry
Community_Discussion	cannot wait, eagles, campaign, fav, doorstep, day is, cast of, decade will, accurate, but
Feature_Feedback	ended the, buffs, be ever, aggressive, consecutive, damn, cameos to, considered, be featured, damn forgot
Technical_Issues	appeal, be over, be my, 35, episode, answered, 31st plenary, and friend, 3310, 31st in

- The Linear SVM identified different TF-IDF features for each topic class.
- Some features are clearly dataset-specific and do not directly represent the semantic meaning of the topic.
- The coefficients represent statistical associations learned from the dataset, rather than universal topic definitions.

8.5 Combined Findings

- Sentiment errors are mainly concentrated around Neutral.
- Topic errors are mainly concentrated around minority topics being classified as Community_Discussion.
- TF-IDF coefficient analysis provides feature-level insight into the behaviour of the final linear models.
- The error patterns and explainability analysis together identify areas where the semantic classification layer remains challenging.

9. Conclusion

The Social Engine's failed semantic understanding layer was rebuilt using an NLP-based classification pipeline on the labelled textual dataset. The study evaluated classical machine learning, lexicon-based sentiment analysis, and lightweight deep learning approaches for sentiment and topic classification.

- TF-IDF + Logistic Regression was selected as the final trainable sentiment model, achieving 57.34% accuracy and 0.5730 macro F1.
- TF-IDF + Linear SVM was selected as the final topic model, achieving 93.49% accuracy and 0.6706 macro F1.
- Group-aware 75:25 splitting prevented identical text entries from appearing in both training and test sets.
- Topic classification showed a substantial improvement over the baseline, with macro F1 increasing from 0.3723 to 0.6706 using the selected topic model.
- Error analysis showed that Neutral sentiment remained the main source of sentiment confusion, while minority topics were frequently classified as Community_Discussion.
- TF-IDF coefficient analysis provided feature-level insight into the classification behaviour of the selected linear models.
- The final models were saved together with the TF-IDF vectorizer, providing a reusable NLP pipeline for future predictions.

10. Limitations

1. Dataset size: The dataset contains 9,000 text records, which limits the amount of labelled data available for learning more complex language patterns.
2. Topic class imbalance: Community_Discussion represents 86.13% of the dataset, while Account_Security represents only 1.51%. This affects minority-class recall and classification performance.
3. Sentiment ambiguity: Sentiment classification remained challenging, particularly for the Neutral class, which was frequently confused with Negative and Positive.
4. Repeated text entries: The dataset contains 1,100 repeated text records. A group-aware split was therefore used to prevent identical cleaned texts from appearing in both training and testing data.
5. TF-IDF limitations: TF-IDF represents text using statistical word and phrase importance and does not capture deeper contextual or semantic relationships between words.
6. Dataset-specific features: Some high-weight TF-IDF features identified during explainability analysis were dataset-specific and may not generalize to other social-media datasets.
7. Model generalization: The reported results are based on the provided Dataset 2 and the fixed test set; performance on unseen datasets or different social-media domains was not evaluated.
8. Lightweight modelling scope: The final pipeline focuses on lightweight classical NLP models. More complex transformer-based approaches were not included in the final evaluation.